

Exam Integrity Digital Transformation Case study

Business Analysis & Digital Transformation

1. Stakeholder Analysis – CATWOE

Customers

Online exam students who are affected by the accuracy and fairness of the monitoring system. Educational institutions are also affected because they need to protect the integrity of their examinations.

Actors

The IT and technical team responsible for maintaining the system, as well as human invigilators who review flagged incidents.

Transformation

Transform an unreliable monitoring system that can wrongly flag honest students and miss suspicious behaviour into a more accurate and fair system that identifies unusual situations while protecting students from unnecessary accusations.

Weltanschauung

The current system can be unreliable. False flags can affect students and may have an impact on their academic progress and wellbeing. At the same time, cheating needs to be detected to protect the credibility of examinations and qualifications.

Owner

The executives or leadership of the exam monitoring company or institution who have the authority to approve and invest in changes to the system.

Environmental Constraints

- Low or unreliable internet bandwidth
- Cost of implementing the solution
- Data privacy and protection
- Different skin tones and lighting conditions
- Different student devices and environments

2. Current State Analysis

The current process requires a student to log into the examination system and enter the required information. The webcam and microphone are then used to monitor the student during the examination.

The system may flag unusual movement, sounds, coughing, standing up, a phone appearing on camera, or other unusual activity.

However, there are some problems with the current process:

Oversensitivity

The system may flag normal behaviour, such as coughing, changing position or briefly leaving the camera view.

Undersensitivity

Some suspicious behaviour may not be detected. For example, a student could potentially use a phone that is outside the camera's view.

Facial Recognition

Facial detection may not work equally well in different lighting conditions or across different skin tones if the system has not been properly trained and tested using diverse data.

Network Problems

High data requirements can cause problems for students with slow or unreliable internet connections. This can interrupt the examination and create an unfair experience.

3. Future State Vision

The proposed future state is a more reliable and fair exam monitoring process.

The system should use multiple signals before creating a significant flag instead of treating every unusual movement as suspicious.

A human invigilator should review important flags before any formal action is taken against the student.

A secondary camera, such as a student's phone, could also provide another angle of the examination environment.

The facial detection system should be tested across different skin tones and lighting conditions.

The system should also use less data and have a process for handling temporary network interruptions so that students are not unfairly affected by connectivity problems.

4. Gap Analysis

Current Gap	Proposed Solution
Normal behaviour can be flagged as suspicious	Use multiple signals and human review
A single camera may miss activity outside its view	Consider a secondary camera

Facial detection may perform differently across conditions	Test across different skin tones and lighting conditions
High data usage can cause network problems	Introduce a low-data mode and network recovery process

5. Requirements

Requirement 1: Multiple Signals and Human Review

Functional Requirement:

As a student, I want the system to consider multiple unusual signals before creating a significant flag so that normal behaviour does not automatically result in an investigation.

Non-Functional Requirement:

The system should notify the appropriate invigilator within the required response time when multiple suspicious signals meet the defined threshold.

Priority: Must Have

Requirement 2: Secondary Camera

Functional Requirement:

The future system should allow a secondary device, such as a mobile phone, to provide an additional camera angle of the examination environment.

Non-Functional Requirement:

The secondary device should connect and begin monitoring within the defined time after the examination starts.

Priority: Must Have

Requirement 3: Facial Detection

Functional Requirement:

The system should detect and verify a student's face under different skin tones and lighting conditions.

Non-Functional Requirement:

The system should be tested using representative data covering different skin tones and lighting conditions before deployment.

Priority: Must Have

Requirement 4: Low-Bandwidth Support

Functional Requirement:

The system should continue supporting the examination process during temporary network problems where technically possible.

Non-Functional Requirement:

The system should operate within a technically defined minimum bandwidth requirement. The exact requirement should be confirmed through technical testing.

Priority: Must Have

6. Process Model

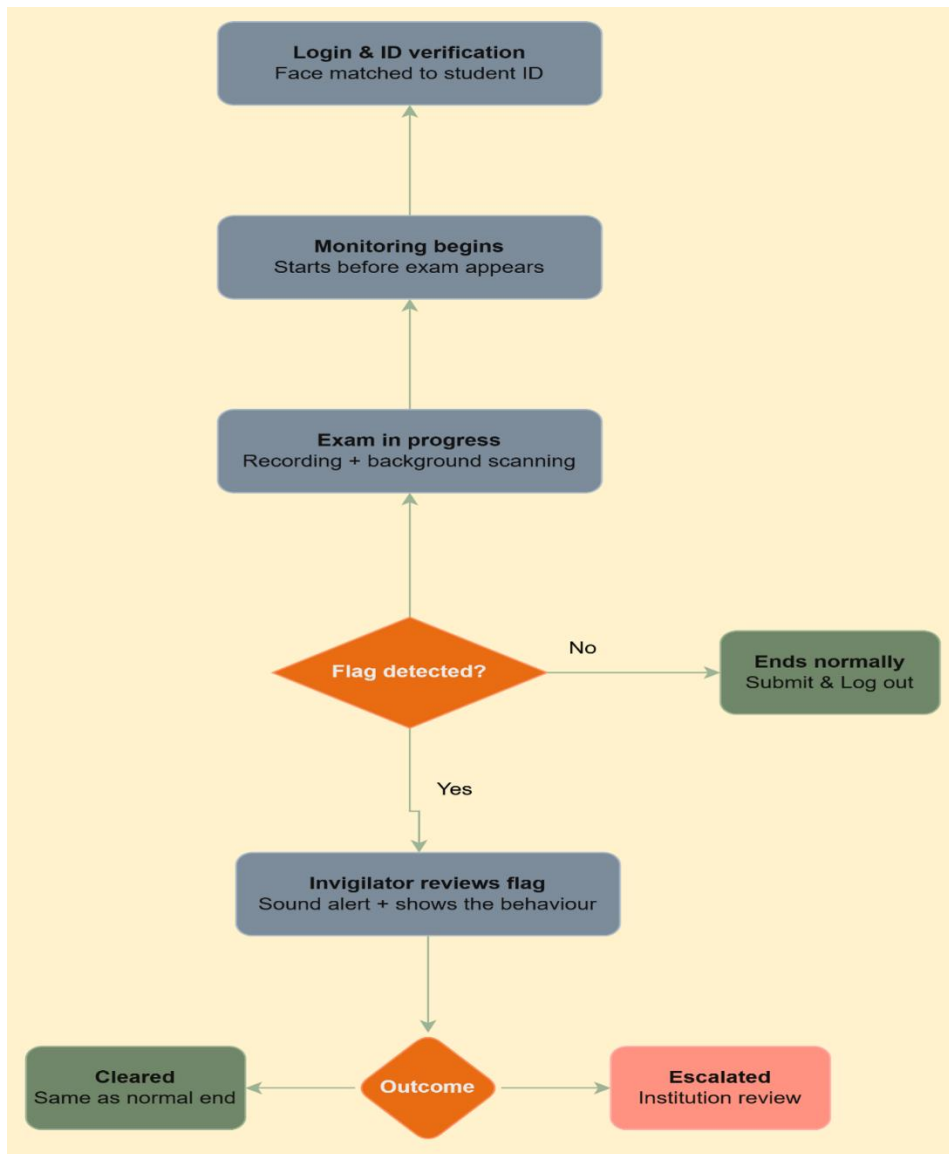
The process model shows the full online examination process from student login to submission.

It includes monitoring the student during the examination and identifying unusual situations.

There are two main decision points:

- Whether an unusual situation should be flagged.
- Whether the human invigilator clears the flag or escalates it for further review.

The process model shows how human review is included before any formal action is taken against the student.



7. Prototype

A basic Python prototype was developed to demonstrate part of the proposed exam monitoring solution.

The prototype demonstrates:

- Webcam monitoring
- Face detection
- No-face detection
- Multiple-face detection
- Unusual incident counting

- Human-review flagging
- Event logging
- Session timing
- End-of-session summary

The prototype does not determine whether a student is cheating. It identifies unusual situations and demonstrates how these situations could be referred for human review.

The prototype is a proof of concept and would require further development and testing before being used in a real examination environment.

8. Change Management

Introducing an improved exam monitoring system will require changes for both students and human invigilators.

Students

Students will need to understand how the new monitoring process works, especially if a secondary camera or other monitoring requirements are introduced. Clear instructions and guidance should be provided before the examination.

Human Invigilators

Invigilators will need training on how to use the new system, review flagged incidents and make decisions based on the available information.

IT and Technical Team

The technical team will need to understand the new system requirements and provide support during implementation. The system should also be tested before being used in real examinations.

Communication

The changes should be communicated clearly to students, invigilators and other relevant stakeholders. This can help reduce confusion and resistance to the new process.

Implementation

The improved system could first be tested with a small group before being introduced more widely. Feedback from students, invigilators and the technical team can then be used to improve the system.

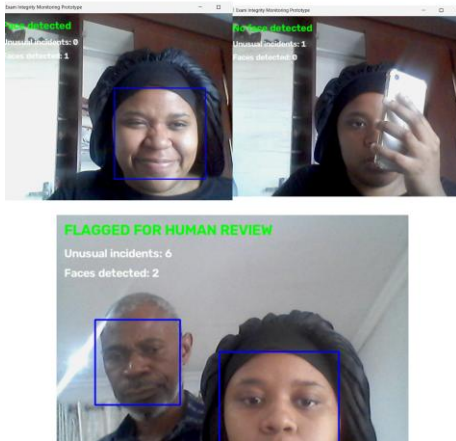
The main purpose of change management is to make sure that people are prepared for the new process and that the technology is successfully adopted.

9. Initial Testing

The prototype was tested using different scenarios to check whether the main functions worked as expected.

Test	Expected Result	Result
One face visible	Face detected	Pass
No face visible	No face detected	Pass
Multiple faces visible	Multiple faces detected	Pass
Unusual situation occurs	Incident counter increases	Pass
Multiple incidents occur	Human-review flag appears	Pass
Session ends	Summary is displayed	Pass

During testing, the incident counter was initially increasing too many times because it was counting every camera frame. The logic was changed so that an incident is counted when the situation changes. After this change, the counter worked as intended.



10. Conclusion

The project identified problems in online exam monitoring, including false flags, missed suspicious activity, facial detection challenges and network problems.

The proposed solution focuses on multiple signals, human review, additional camera coverage, improved facial detection testing and better support for unreliable internet connections.

The Python prototype demonstrates some of these ideas in a working proof of concept and shows how a Business Analyst can move from identifying a business problem to analysing the current state, defining requirements, designing a future state and working with technology to test a possible solution.

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Self-Directed Portfolio Project

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